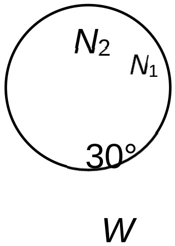
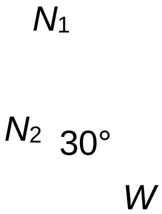
6 **Ans: C**





Resolving the forces,

$N_2 \sin 30 = N_1 \quad (1)$

$N_2 \cos 30 = W \quad (2)$

$\frac{(1)}{(2)}: \qquad \tan 30 = \frac{N_1}{W}$

$N_1 = W \tan 30 = 15 \tan 30 = 8.66 = 8.7$ N (2s.f.)

(Alternative: Using vector triangle)

$N_1 = W \tan 30 = 15 \tan 30 = 8.66 = 8.7$ N (2s.f.)

7 **Ans: A**